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# Soil and Water Conservation Engineering

BOE078

One-day revision notes: definitions, solved examples, code and important questions

**Program:** B.Tech | **Branches:** ALL | **Semester:** 7 | **Credits:** 3 | **L-T-P:** 3-0-0

**Scheme:** AKTU B.Tech 4th year, Open Elective-II, 2025-26 onwards


**UNIT 1** Soil Erosion, USLE and Wind Erosion

**UNIT 2** Water Erosion and Engineering Control Measures

**UNIT 3** Biological Control, Soil Management and Water Conservation

**UNIT 4** Water Losses and Agricultural Water Pollution

**UNIT 5** Water Resource Planning and Flood Management


Original revision notes written for the AKTU syllabus. Every unit has the same parts:

1. **2-mark definitions**: short answers you can write word for word in Section A.
2. **Core concepts**: processes, structures and comparison tables.
3. **Solved examples**: the numericals that appear in the paper (USLE soil loss, contour bund spacing, bench terrace design, rational method runoff, SCS curve number, reservoir sedimentation, rainwater harvesting, Darcy's law, groundwater recharge, benefit-cost ratio).
4. **Formulas / key points to remember**.
5. **Important questions**: the long questions to prepare, with what your answer must contain.

## How to use these notes the day before the paper

- **Morning (3 hours):** Unit 1 (soil conservation, causes and mechanics of erosion, USLE, wind erosion and control, vegetative practices) and Unit 2 (sheet, rill and gully erosion, sediment transport and reservoir sedimentation, bunds, terraces, waterways, spillways, check dams, river training).
- **Afternoon (3 hours):** Unit 3 (biological control, grassland and forest management, soil quality, drainage, salt-affected soils, water harvesting, rainfall-runoff, storage, groundwater recharge) and Unit 4 (infiltration, seepage and evaporation losses, water pollution from agriculture, self-purification, pollutant movement in groundwater).
- **Evening (2 hours):** Unit 5 (planned water resource use, economics, flood plain management, flood modification and damage reduction). Then the "Last-hour checklist".
- **In the exam:** draw neat cross-sections of contour bunds, bench terraces, a drop spillway, a check dam and a farm pond, and write the formula before each calculation.

**AKTU paper pattern (theory, 70 marks, 3 hours):** Section A has 7 short questions of 2 marks, all compulsory (14). Section B has 5 questions of 7 marks, attempt any 3 (21). Section C has 5 questions

(one per unit), each with two parts, attempt one part of each, 7 marks each (35). Every unit gets its own Section C question, so do not skip any unit.

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# UNIT 1 Soil Erosion, USLE and Wind Erosion

## ✍️ 2-mark definitions

- **Soil conservation:** the use of land within its capability and the application of practices that protect soil from erosion and maintain its fertility and productivity.
- **Soil erosion:** the detachment and transport of soil particles from one place to another by water, wind, ice or gravity.
- **Geological (natural) erosion:** slow erosion under natural vegetation, roughly balanced by soil formation.
- **Accelerated erosion:** erosion faster than soil formation, caused by human activities like deforestation, overgrazing and poor farming.
- **Detachment and transport:** the two phases of erosion; raindrop impact or wind detaches particles, runoff or wind carries them.
- **Universal Soil Loss Equation (USLE):** an empirical equation (Wischmeier and Smith) that estimates average annual soil loss from sheet and rill erosion: A = R K L S C P.
- **Erosivity (R):** the ability of rainfall to cause erosion, related to rainfall energy and maximum 30-minute intensity ($EI_{30}$).
- **Erodibility (K):** the susceptibility of a soil to erosion, depending on texture, organic matter, structure and permeability.
- **Saltation:** wind erosion movement where particles of 0.1–0.5 mm bounce along the surface (50–75% of movement).
- **Suspension:** fine particles under 0.1 mm carried high in the air (3–40%).
- **Surface creep:** larger particles 0.5–1 mm rolled along the surface by wind and saltating particles (5–25%).
- **Strip cropping:** growing erosion-permitting crops (maize, sorghum) and erosion-resisting crops (legumes, grasses) in alternate strips.
- **Stubble mulching:** leaving crop residues on the surface after harvest to protect the soil.

## 💡 Core concepts

**Scope of soil conservation:** protects agricultural productivity, reduces siltation of reservoirs and rivers, improves groundwater recharge, prevents floods and droughts, sustains forests and pastures, protects infrastructure, supports livelihoods. In India about a third of the land area is degraded; the annual soil loss is estimated in billions of tonnes.

**Causes of soil erosion:** deforestation, overgrazing, shifting cultivation, up-and-down slope cultivation, improper irrigation, steep slopes, intense rainfall, high wind, mining and construction, lack of crop cover, fire.

**Mechanics of water erosion:** (1) raindrop impact detaches particles (splash erosion), seals the surface and reduces infiltration; (2) runoff forms and carries detached soil; (3) concentrated flow increases detachment, forming rills and gullies; (4) deposition when velocity falls.

**Universal Soil Loss Equation: A = R × K × L × S × C × P**

| Factor | Meaning |
|---|---|
| A | Average annual soil loss (t/ha/year) |
| R | Rainfall erosivity factor |
| K | Soil erodibility factor |
| L | Slope length factor (relative to 22.1 m) |
| S | Slope steepness factor (relative to 9%) |
| C | Cover and management factor (1 for bare fallow, lower with crops) |
| P | Support practice factor (1 for up-and-down slope; lower for contouring, strip cropping, terracing) |

Uses: predict soil loss, compare practices, plan conservation to keep loss below the tolerable limit (T, about 5–12 t/ha/year). Limitations: empirical, average annual values only, ignores gully erosion and deposition, single storms; modified versions RUSLE and MUSLE.

**Wind erosion:** occurs in arid and semi-arid areas (Rajasthan, Gujarat) with loose, dry, fine soil, little cover and strong winds (threshold about 5 m/s at 30 cm height). Effects: loss of topsoil, sand dune movement, crop damage, dust storms, health problems.

**Wind erosion control**

| Method | How it helps |
|---|---|
| Windbreaks and shelterbelts | Rows of trees perpendicular to wind reduce speed for a distance of about 10–20 times their height downwind |
| Strip cropping across wind | Strips of resistant crops trap moving soil |
| Stubble mulching and crop residue | Covers soil, increases surface roughness |
| Cover crops | Keep the soil covered in off-season |
| Tillage practices | Rough, cloddy surface; tillage across the wind; minimum tillage |
| Sand dune stabilization | Micro-windbreaks, grasses (Sewan, Lasiurus), tree planting |
| Irrigation | Moist soil resists wind |

**Vegetation management:** contour cropping, strip cropping (contour, field and buffer strip), cover crops, crop rotation with legumes, mulching (stubble, plastic, straw), agroforestry, grassed waterways.

## 🧮 Solved examples

**Example 1: USLE.** R = 500, K = 0.3, LS = 1.2, C = 0.4 (maize), P = 0.5 (contour strip cropping).

- A = 500 × 0.3 × 1.2 × 0.4 × 0.5 = **36 t/ha/year**.
- With up-and-down slope farming (P = 1.0) the loss would be 72 t/ha/year; the support practice halves it. If the tolerable loss is 12 t/ha/year, terracing (P ≈ 0.15) would give 36 × 0.15/0.5 = 10.8 t/ha/year, acceptable.

**Example 2: Windbreak protection.** A shelterbelt 10 m high protects a distance of about 10 to 20 times its height on the leeward side, so 100–200 m. For a field 800 m long in the wind direction, about 4–8 parallel belts are needed.

## Key points to remember

- Erosion = detachment + transport (+ deposition).
- USLE A = RKLSCP; tolerable loss T about 5–12 t/ha/year.
- Wind erosion: saltation (most), suspension, surface creep.
- Wind control: windbreaks, strip cropping across wind, stubble mulching, cover crops, rough tillage.

## ⭐ Important questions

1. Define soil conservation and explain its scope. Explain the causes and mechanics of soil erosion.
2. Explain the Universal Soil Loss Equation with its factors. Solve a numerical.
3. Explain wind erosion, its mechanics and control measures.
4. Explain vegetative management practices: strip cropping, stubble mulching and others.

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# UNIT 2 Water Erosion and Engineering Control Measures

## ✍️ 2-mark definitions

- **Sheet erosion:** the uniform removal of a thin layer of soil from the land surface by overland flow; hard to notice.
- **Rill erosion:** erosion in small channels (a few cm deep) that can be removed by normal tillage.
- **Gully erosion:** advanced erosion forming deep channels that cannot be removed by normal tillage.
- **Sediment transport:** the movement of eroded particles in a channel as bed load (rolling and sliding) and suspended load.
- **Reservoir sedimentation:** deposition of sediment in a reservoir, reducing its storage capacity and useful life.
- **Trap efficiency:** the percentage of incoming sediment that is retained in a reservoir.
- **Contour bund:** an earthen embankment built along the contour to intercept runoff and reduce slope length.
- **Graded bund:** a bund with a slight longitudinal gradient to drain excess water safely.
- **Bench terrace:** a series of level or nearly level platforms (benches) cut across a steep slope, with risers between them.
- **Grassed (vegetated) waterway:** a natural or constructed channel lined with grass to carry runoff safely.
- **Chute spillway:** a steep open channel structure that carries water down a large drop.
- **Drop spillway:** a structure where water falls vertically over a weir into a stilling basin.
- **Drop inlet spillway:** water enters a vertical riser and passes through a conduit under an embankment.
- **Check dam:** a small barrier across a gully or stream to reduce flow velocity, trap sediment and store water.
- **River training:** works that guide a river's flow to protect banks, prevent shifting and maintain a channel (spurs, guide bunds, revetments).

## 💡 Core concepts

**Stages of gully development:** (1) formation (channel erosion by downward scour), (2) development (head cutting upstream and enlargement), (3) healing (vegetation starts), (4) stabilization (reaches stable gradient, vegetation established). Gully classification by depth: small (under 1 m), medium (1–5 m), large (over 5 m).

**Factors in sediment transport:** flow velocity and depth, slope, particle size and density, sediment supply, channel roughness. Bed load and suspended load are estimated by formulas such as Du Boys or Einstein (for bed load) and sampling for suspended load.

**Reservoir sedimentation:** sediment deposits as delta at the head, then spreads; reduces dead and live storage. Control: catchment treatment (most effective), check dams upstream, vegetative screens,

sluicing and flushing during floods, desilting, design with extra dead storage.

**Erosion control measures on agricultural land**

| Measure | Suitable slope | Notes |
|---|---|---|
| Contour bunds | Up to 6% in low-rainfall areas | Retain water (moisture conservation) |
| Graded bunds | 2–6% in high-rainfall areas | Drain excess runoff safely |
| Bench terraces | 16–33% | Convert steep slopes to steps for cultivation |
| Contour trenches | Steep non-agricultural land | Trap runoff for tree plantation |
| Grassed waterways | Outlets for terraces and bunds | Carry runoff safely |

**Bench terrace types:** level (for irrigated paddy), outward sloping (low rainfall), inward sloping (high rainfall, with drain at inner edge), puertorican (gradual formation).

**Gully control structures:** temporary structures (brushwood dams, loose rock dams, log check dams, woven wire dams) for small gullies; permanent structures (drop spillway, chute spillway, drop inlet spillway, masonry check dams) for large flows.

**Drop spillway parts:** headwall and wingwalls, weir (crest), apron (stilling basin) with end sill, sidewalls and toe walls; for drops up to about 3 m. **Chute spillway:** inlet, steep chute and energy dissipator (stilling basin); for drops over 3 m. **Drop inlet:** riser and pipe through an earth dam; used to control gully heads and in farm ponds.

**River training works:** spurs or groynes (project from the bank to deflect flow), guide bunds (guide river through a bridge), revetments and pitching (protect banks with stones), levees and embankments, cut-offs, and bank protection with vegetation.

## 🧮 Solved examples

**Example 1: Contour bund spacing.** Land slope 6%. Using vertical interval VI = 0.3 (S/a + b) with a = 3 and b = 2 (low-rainfall, light soil):

- VI = 0.3 (6/3 + 2) = 0.3 × 4 = **1.2 m**.
- Horizontal interval HI = VI × 100 / S = 1.2 × 100 / 6 = **20 m** between bunds.

**Example 2: Bench terrace depth of cut.** Land slope S = 20%, bench width W = 5 m, riser batter 1:1 (U = 1). Vertical interval D = S W / (100 − S U) = 20 × 5 / (100 − 20) = **1.25 m**. Depth of cut ≈ D/2 = 0.625 m.

**Example 3: Reservoir life.** A reservoir of 100 $Mm^3$ receives 1 $Mm^3$ of sediment per year with 90% trap efficiency. Annual loss = 0.9 $Mm^3$. Time to lose 50% of capacity = 50 / 0.9 = **55.6 years**.

**Example 4: Grassed waterway capacity.** A parabolic waterway with area 1.5 $m^2$ and safe velocity 1.5 m/s (good grass) carries Q = A V = **2.25 $m^3/s$**.

## Key points to remember

- Water erosion stages: splash → sheet → rill → gully (→ stream bank).
- Gully stages: formation, development, healing, stabilization.

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- Contour bunds (low rainfall), graded bunds (high rainfall), bench terraces (steep slopes).
- VI = 0.3(S/a + b), HI = 100 VI/S; bench terrace D = SW/(100 − SU).
- Drop spillway up to 3 m; chute over 3 m; drop inlet through embankments.

## ⭐ Important questions

1. Explain the types of water erosion: sheet, rill and gully. Explain the stages of gully development.
2. Explain sediment transport in channels and reservoir sedimentation and its control.
3. Explain bunding and terracing. Design contour bunds and bench terraces.
4. Explain vegetated waterways, chute spillways, drop inlet spillways and check dams.
5. Explain river training works.

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# UNIT 3 Biological Control, Soil Management and Water Conservation

## ✍️ 2-mark definitions

- **Biological (agronomic) measures:** using vegetation and crop management to control erosion: cover crops, mulching, contour farming, strip cropping, agroforestry.
- **Grassland management:** managing pastures to keep good grass cover through controlled grazing, reseeding and fertilization.
- **Forest management:** managing forests for protection and production through afforestation, reforestation, fire control and controlled felling.
- **Soil quality:** the capacity of a soil to function: sustain plant growth, regulate water, filter pollutants and support biodiversity.
- **Drainage:** removal of excess surface or subsurface water from land to improve crop growth.
- **Saline soil:** soil with high soluble salts (EC above 4 dS/m), pH below 8.5, ESP below 15.
- **Sodic (alkali) soil:** soil with high exchangeable sodium (ESP above 15), pH above 8.5, poor structure.
- **Water harvesting:** collecting and storing rainfall runoff for later use or recharge.
- **Runoff coefficient (C):** the ratio of runoff to rainfall.
- **Groundwater recharge:** replenishment of aquifers by water percolating from the surface, naturally or by artificial structures.

## 💡 Core concepts

**Biological control methods:** contour farming, strip cropping, cover crops, mulching, crop rotation, mixed cropping, agroforestry (trees with crops), vegetative barriers (vetiver grass strips), afforestation on steep slopes.

**Grassland management:** rotational and deferred grazing, limiting stocking rate to carrying capacity, reseeding with good species, fertilizing, controlling weeds and fire, contour furrows to hold water.

**Forest management:** afforestation of wastelands, protection from fire, grazing and illegal felling, social forestry and joint forest management, selective felling instead of clear-felling, catchment area treatment.

**Soil quality management:** add organic matter (compost, green manure), balanced fertilization, conservation tillage, crop rotation with legumes, avoid compaction, maintain pH (lime for acid soils, gypsum for sodic soils).

**Drainage works:** surface drainage (open field ditches, land grading) and subsurface drainage (tile or perforated pipe drains, mole drains) that lower the water table; vertical drainage by tube wells. Benefits: aeration, prevents waterlogging and salinity, earlier tillage.

**Reclamation of salt-affected soils**

| Soil | Reclamation |
|---|---|
| Saline | Leaching with good-quality water plus subsurface drainage; salt-tolerant crops |
| Sodic | Gypsum (or sulphur, pyrite) to replace sodium with calcium, then leaching; organic matter; rice and dhaincha green manure |
| Saline-sodic | Gypsum first, then leaching (leaching alone makes it sodic) |

**Water conservation and harvesting**

- In situ: contour farming, ridges and furrows, broad bed and furrow, mulching, conservation tillage.
- Ex situ: farm ponds, tanks, check dams, percolation tanks, rooftop harvesting, khadins, johads.
- **Rainfall-runoff relation:** runoff depends on rainfall intensity and duration, soil infiltration, land use, slope and antecedent moisture. **Rational method:** $Q = C\,I\,A / 360$ ($Q$ in $m^3/s$, $I$ in mm/h, $A$ in ha). **SCS curve number method:** $Q = (P - I_a)^2 / (P - I_a + S)$, $S = 25{,}400/CN - 254$ (mm), $I_a = 0.2\,S$.

**Water storage:** ponds (dugout, embankment), lakes, reservoirs; design considers storage capacity, evaporation and seepage losses, spillway capacity; aquifers as underground storage (no evaporation loss).

**Groundwater recharge methods:** percolation tanks, check dams, recharge wells and shafts, dug well recharge, recharge trenches, injection wells, spreading basins, subsurface dykes; managed aquifer recharge.

## 🧮 Solved examples

**Example 1: Rational method.** Catchment 20 ha, runoff coefficient 0.4, rainfall intensity 60 mm/h for the time of concentration. $Q = 0.4 \times 60 \times 20 / 360 =$ **1.33 $m^3/s$**. A waterway or spillway must carry this peak.

**Example 2: SCS curve number.** CN = 80, rainfall P = 100 mm.

- $S = 25{,}400/80 - 254 = 63.5$ mm; $I_a = 0.2 \times 63.5 = 12.7$ mm.
- $Q = (100 - 12.7)^2 / (100 - 12.7 + 63.5) = 7621 / 150.8 =$ **50.5 mm** of runoff.

**Example 3: Rooftop harvesting.** Roof area 100 $m^2$, annual rainfall 800 mm, collection efficiency 0.8. Water harvested = $100 \times 0.8 \times 0.8 =$ **64 $m^3$** (64,000 litres) per year.

**Example 4: Recharge.** A 10 ha watershed with 800 mm rain and 15% recharge with structures gets $10 \times 10^4 \times 0.8 \times 0.15 =$ **12,000 $m^3$** of recharge per year.

**Example 5: Gypsum requirement (principle).** Gypsum ($CaSO_4 \cdot 2H_2O$) supplies $Ca^{2+}$, which replaces exchangeable $Na^+$ on clay; the sodium sulphate formed is leached away with drainage water. The dose depends on how much exchangeable sodium must be replaced (from the soil's CEC and the drop in ESP wanted), the soil depth treated and its bulk density, and is fixed by a laboratory gypsum requirement test. One milliequivalent of Ca per 100 g of soil over 15 cm depth needs about 0.86 g of gypsum per kg of soil; with 1.95–2.25 million kg of soil per hectare (bulk density 1.3–1.5 g/$cm^3$), that is about 1.7–1.9 t/ha of pure gypsum.

## Key points to remember

- Biological control: cover, mulching, contour farming, strip cropping, agroforestry, vetiver.
- Saline vs sodic soils (EC, pH, ESP); gypsum for sodic.
- Surface and subsurface drainage.
- Rational Q = CIA/360; SCS-CN Q = $(P - 0.2S)^2/(P + 0.8S)$.
- Harvesting: in situ and ex situ; recharge structures.

## ⭐ Important questions

1. Explain biological methods of erosion control, grassland and forest management.
2. Explain soil quality management and drainage works.
3. Explain the reclamation of saline and sodic soils.
4. Explain water conservation and harvesting. Explain the rainfall-runoff relationship with the rational and SCS methods.
5. Explain water storage in ponds, lakes, reservoirs and aquifers, and methods of groundwater recharge.

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# UNIT 4 Water Losses and Agricultural Water Pollution

## ✍️ 2-mark definitions

- **Infiltration:** entry of water into the soil surface.
- **Percolation:** downward movement of water through the soil below the root zone.
- **Seepage:** loss of water through the bed and sides of canals, ponds and reservoirs.
- **Evaporation:** conversion of water from a free surface to vapour.
- **Evapotranspiration:** the combined loss by evaporation from soil and transpiration from plants.
- **Point source pollution:** pollution from a single identifiable source, like a factory outlet.
- **Non-point source pollution:** diffuse pollution from large areas, like fertilizer and pesticide runoff from farms.
- **Eutrophication:** excessive nutrient (N, P) enrichment of water bodies causing algal blooms and oxygen depletion.
- **BOD (Biochemical Oxygen Demand):** the oxygen used by microbes to decompose organic matter in water, a measure of organic pollution.
- **Self-purification:** the natural process by which a stream recovers its quality through dilution, sedimentation, oxidation, aeration and biological decay.
- **Dispersion:** spreading of a pollutant in groundwater due to velocity variations and diffusion as it moves.
- **Advection:** transport of a pollutant with the bulk flow of groundwater.

## 💡 Core concepts

### Water losses and their control

| Loss | Control |
|---|---|
| Infiltration and deep percolation (excess irrigation) | Efficient irrigation (drip, sprinkler), land levelling, proper scheduling |
| Seepage from canals and ponds | Lining (concrete, brick, plastic film, bentonite, compacted clay) |
| Evaporation from reservoirs | Windbreaks, reducing surface area, chemical films (cetyl alcohol), floating covers, underground storage |
| Evapotranspiration | Mulching, weed control, drought-tolerant crops |

**Infiltration rate:** highest at the start, decreasing to a steady rate; measured by double-ring infiltrometer; Horton's equation f = f_c + ($f_o$ − f_c) e^(−kt).

**Darcy's law:** flow through porous media Q = K i A, where K is hydraulic conductivity and i the hydraulic gradient.

**Sources of agricultural water pollution:** fertilizers (nitrate leaching to groundwater, phosphate runoff causing eutrophication), pesticides (persistent organochlorines, toxicity), animal wastes (pathogens,

BOD, nutrients), sediment (turbidity, carries P and pesticides), salts from irrigation return flow, crop residue burning effects, agro-industrial effluents.

**Effects:** nitrate above 45 mg/L (as $NO_3$) in drinking water causes methaemoglobinaemia (blue baby syndrome); pesticide bioaccumulation and biomagnification; fish kills; algal blooms; loss of biodiversity.

**Control:** balanced and split fertilizer application, slow-release fertilizers, integrated pest management, buffer strips and constructed wetlands, proper manure management, conservation tillage, efficient irrigation to reduce leaching.

**Self-purification of surface water:** physical (dilution, sedimentation, aeration from the atmosphere, sunlight), chemical (oxidation, reduction), biological (bacterial decomposition of organic matter). **Oxygen sag curve (Streeter-Phelps):** downstream of a waste outfall, dissolved oxygen drops as BOD is consumed (deoxygenation), reaches a minimum (critical point), then recovers through reaeration. Zones: degradation, active decomposition, recovery, clear water.

**Pollutant dispersion in groundwater:** processes: advection (with flow), mechanical dispersion and molecular diffusion (spreading), sorption (retardation by soil), degradation (biological and chemical decay). A pollutant plume spreads and elongates in the flow direction; clay and organic matter retard pesticides; nitrate moves freely.

## 🧮 Solved examples

**Example 1: Darcy's law.** Aquifer cross-section 500 $m^2$, hydraulic conductivity 10 m/day, hydraulic gradient 0.01. Q = 10 × 0.01 × 500 = **50 $m^3$/day**.

**Example 2: Canal seepage saving.** An unlined canal 5 km long with wetted perimeter 6 m loses 0.3 $m^3$ per $m^2$ per day. Wetted area = 5000 × 6 = 30,000 $m^2$, loss = 9,000 $m^3$/day. Lining reducing seepage by 80% saves **7,200 $m^3$/day**.

**Example 3: Reservoir evaporation.** A pond with surface area 2 ha and pan evaporation 6 mm/day (pan coefficient 0.7) loses $2 \times 10^4 \times 0.006 \times 0.7$ = **84 $m^3$/day**.

**Example 4: Nitrate check.** Groundwater has 13 mg/L nitrate-nitrogen ($NO_3$-N). As nitrate: 13 × 62/14 = 57.6 mg/L, above the 45 mg/L limit, so unsafe for drinking.

## Key points to remember

- Losses: infiltration/percolation, seepage, evaporation; controls for each.
- Darcy Q = KiA; Horton infiltration.
- Agricultural pollution: nitrate, phosphate, pesticides, sediment, animal waste, salts.
- Self-purification: dilution, sedimentation, oxidation, reaeration; oxygen sag curve.
- Groundwater pollutant transport: advection, dispersion, diffusion, sorption, decay.

## ⭐ Important questions

1. Explain water losses: infiltration, seepage and evaporation, and their control.
2. Explain water quality pollution from fertilizers and pesticides.
3. Explain the sources of agricultural water pollution and control measures.

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4. Explain the self-purification of surface water with the oxygen sag curve.
5. Explain the dispersion of pollutants in groundwater.

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# UNIT 5 Water Resource Planning and Flood Management

## ✍️ 2-mark definitions

- **Water resources planning:** the systematic process of developing, distributing and managing water to meet present and future needs sustainably.
- **Integrated water resources management (IWRM):** coordinated management of water, land and related resources to maximize welfare without compromising ecosystems.
- **Watershed:** the area draining to a common outlet; the natural planning unit for soil and water conservation.
- **Benefit-cost ratio (BCR):** the ratio of the present worth of benefits to the present worth of costs; a project is viable if BCR is more than 1.
- **Flood:** an overflow of water that submerges land that is usually dry.
- **Flood plain:** the flat land beside a river that is flooded during high flows.
- **Flood plain zoning:** regulating land use in flood plains according to flood risk.
- **Flood forecasting:** predicting the timing and magnitude of floods for warnings.
- **Return period:** the average interval (in years) between floods of a given size or greater; T = 1/P.

## 💡 Core concepts

**Need for planned utilization of water resources:** rising population and demand (agriculture uses about 80% in India), uneven distribution in space and time (monsoon), groundwater depletion, pollution, conflicts between states and sectors, climate change, need for food security and drinking water, and environmental flows.

**Planning principles:** watershed or river basin as the unit, multi-purpose projects (irrigation, power, drinking water, flood control), conjunctive use of surface and groundwater, demand management (pricing, efficient irrigation), participation of users, environmental sustainability, priorities as in the National Water Policy (drinking water first).

**Economics of water resource utilization:** costs (capital, operation and maintenance, environmental and resettlement), benefits (crop production, hydropower, flood damage avoided, drinking water, fisheries); evaluation by BCR, NPV and IRR; water pricing for cost recovery and conservation; social and environmental cost-benefit.

**Flood management approaches**

| Approach | Measures |
|---|---|
| Modifying the flood (structural) | Dams and reservoirs (flood storage), detention basins, embankments and levees, channel improvement, diversion channels, watershed treatment (afforestation, check dams) |
| Reducing susceptibility to damage (non-structural) | Flood plain zoning, building codes, flood proofing (raised plinths, stilts), flood forecasting and warning, land-use regulation, relocation |

| Approach | Measures |
|---|---|
| Reducing the impact (mitigating losses) | Disaster preparedness, evacuation, relief and rehabilitation, flood insurance, emergency services, community awareness |

**Flood plain zone management:** prohibited zone (floodway, frequent floods, only parks and agriculture), restricted zone (moderate risk, buildings with raised floors), warning zone (rare floods, normal use with warnings). In India, Central Water Commission guidelines suggest zones based on 10-, 25- and 100-year floods.

**Flood forecasting:** rainfall-runoff models, gauge-to-gauge correlation, real-time telemetry, satellite rainfall; warnings to districts; Central Water Commission runs India's flood forecasting network.

## 🧮 Solved examples

**Example 1: Benefit-cost ratio.** A check dam costs ₹10 lakh with ₹50,000 per year maintenance and gives benefits of ₹2.5 lakh per year for 15 years. At 10%, annuity factor = 7.606. PV of benefits = 2.5 × 7.606 = ₹19.0 lakh; PV of costs = 10 + 0.5 × 7.606 = ₹13.8 lakh. **BCR = 19.0 / 13.8 = 1.38**, viable.

**Example 2: Return period.** A flood with annual exceedance probability 0.01 has a return period of **100 years**. The chance of at least one such flood in a 30-year period = $1 - (0.99)^{30}$ = **0.26 (26%)**, which is why 100-year floods matter for planning.

**Example 3: Detention storage.** A detention pond must reduce a flood peak by storing 2 $m^3/s$ of excess inflow for 3 hours. Storage = 2 × 3 × 3600 = **21,600 $m^3$**.

## Key points to remember

- Need for planning: demand, variability, depletion, pollution, conflicts.
- Economics: BCR more than 1, NPV, water pricing.
- Flood management: modify the flood (structural), reduce susceptibility (zoning, flood proofing, warning), reduce impact (relief, insurance).
- Return period T = 1/P; risk in n years = $1 - (1 - 1/T)^n$.

## ⭐ Important questions

1. Explain the need for planned utilization of water resources.
2. Explain the economics of water resource utilization.
3. Explain flood plain zone management.
4. Explain measures for modifying floods, reducing susceptibility to damage, and reducing the impact of flooding.

## Last-hour checklist

- ☐ Soil conservation scope; causes and mechanics of erosion.
- ☐ USLE factors and example (36 t/ha/yr); tolerable loss.

☐ Wind erosion: saltation, suspension, creep; control measures; windbreak protection distance.

☐ Sheet, rill, gully; gully stages; sediment transport; reservoir life (55.6 years).

☐ Contour bund (VI 1.2 m, HI 20 m); bench terrace (D 1.25 m); waterways, chute, drop and drop inlet spillways, check dams, river training.

☐ Biological measures; grassland and forest management; soil quality; drainage; saline vs sodic.

☐ Rational (1.33 $m^3/s$) and SCS-CN (50.5 mm); rooftop harvesting (64 $m^3$); recharge.

☐ Water losses and control; Darcy (50 $m^3$/day); agricultural pollution; self-purification and oxygen sag; groundwater dispersion.

☐ Planned water use; BCR (1.38); flood plain zoning; flood modification and damage reduction; return period risk (26%).

## Books prescribed by AKTU

These are the books named in the official AKTU syllabus. Buy or borrow them through a bookshop or your college library. The WorldCat links show which libraries hold a copy. The syllabus lists titles and authors only; publishers below are the usual Indian editions.

| # | Title | Author(s) | Publisher | Where to get it legally |
|---|---|---|---|---|
| 1 | Modern Geotechnical Engineering | Alam Singh | CBS Publishers | WorldCat |
| 2 | Soil Mechanics and Foundation Engineering | K. R. Arora | Standard Publishers Distributors | WorldCat |
| 3 | The Nature and Properties of Soils | Nyle C. Brady, Ray R. Weil | Pearson | WorldCat |
| 4 | Soil Mechanics and Foundations | B. C. Punmia, Ashok Kumar Jain, Arun Kumar Jain | Laxmi Publications | WorldCat |

The syllabus lists Brady's book as "Principles of Soil Sciences"; it is best known as "The Nature and Properties of Soils". For the conservation topics, V. V. Dhruva Narayana's "Soil and Water Conservation Research in India" (ICAR) and R. Suresh's "Soil and Water Conservation Engineering" (Standard Publishers) are widely used companions.

**Syllabus source:** AKTU Open Elective syllabus for B.Tech final year, 2025-26 (copy hosted by an affiliated college, PDF)